ME 415 - Flight Vehicle Design

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SUBJECT: Milestone #4 - Wing Design
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The goal of our wing design was to maximize stability, span efficiency, and minimize drag. We began our wing design process by using our airfoil design from Milestone 3 as the basis for our wing. By starting with an optimized airfoil, we were able to design a stable and effective wing while considering other challenges such as manufacturability and failure modes.

Our design will have a carbon-spar reinforced polystyrene central wing sandwiched between two wingtips 3D printed in foaming PLA. The printed wingtips will have holes that will mount onto the carbon wing spar. The whole wing will be designed to shear from the body of the aircraft in the event of a crash or impact, and the wingtips will be designed to safely break off of the polystyrene main wing's strong spar (we will design the wingtip to be much weaker than the spar to protect the spar). The wingtips are designed to break to protect the rest of the system, and by 3D printing them, we will be able to rapidly produce spares and take advantage of more complicated geometry in the tips.

Table 1: Central Span Characteristics (Station 0)

Station	**Foil**	**Material**	**Span** (in)	**Chord root** (in)	**Chord tip** (in)	**Taper Ratio**	**Dihedral** (°)	**Twist** (°)
0	e168	Polystyrene	24	7.9	7.9	1	0	0
1	e168	Foaming PLA	10	7.9	7.85	.99	18.4	0.1

To validate that our proposed wing design will be a viable option, we ran a genetic optimization algorithm that uses analyses from FlightLab as follows:

$maximize$: $MAC,\ span\ efficiency,\ pitch\ control\ deflection,\ min\ stall\ speed$

by vary: main wing chord, dihedral wing section taper, dihedral tip elevation, tip twist, airfoil design, tail chord, tail twist, and CG location

$Subject\ to$: $station\ span\ location,\ horizontal\ tail\ station\ location$

Figure 1. Station model of planned wing design

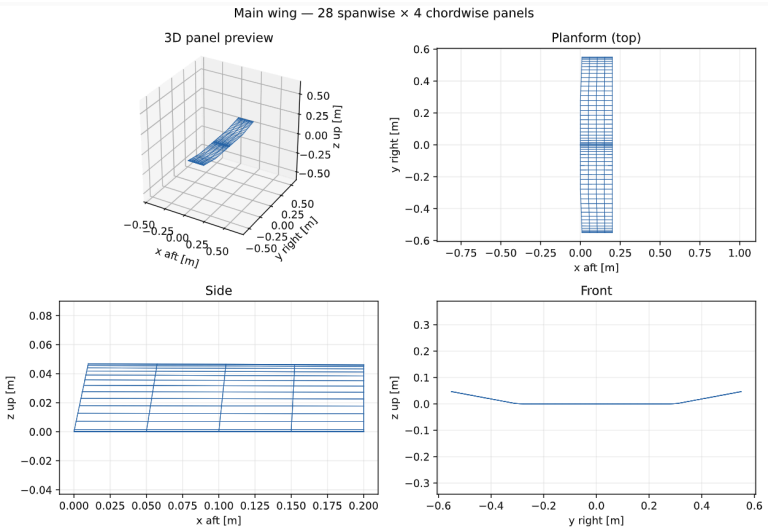


Figure 2. OnShape CAD of proposed wing design

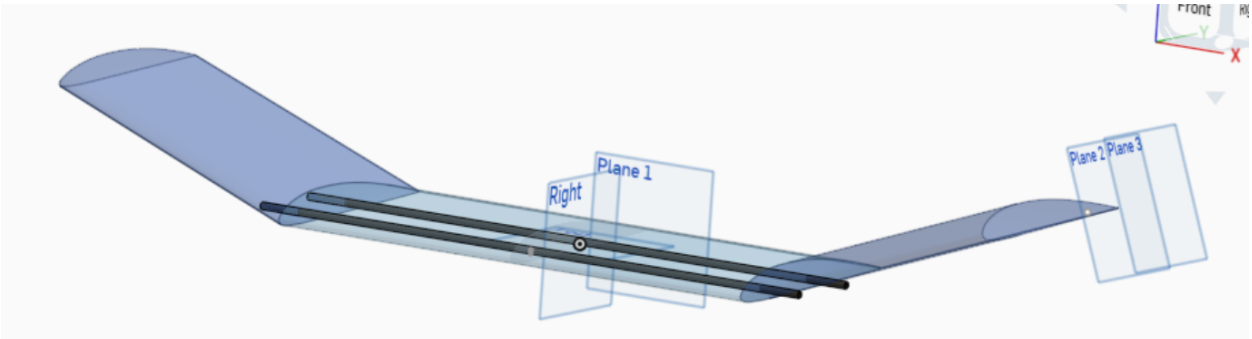
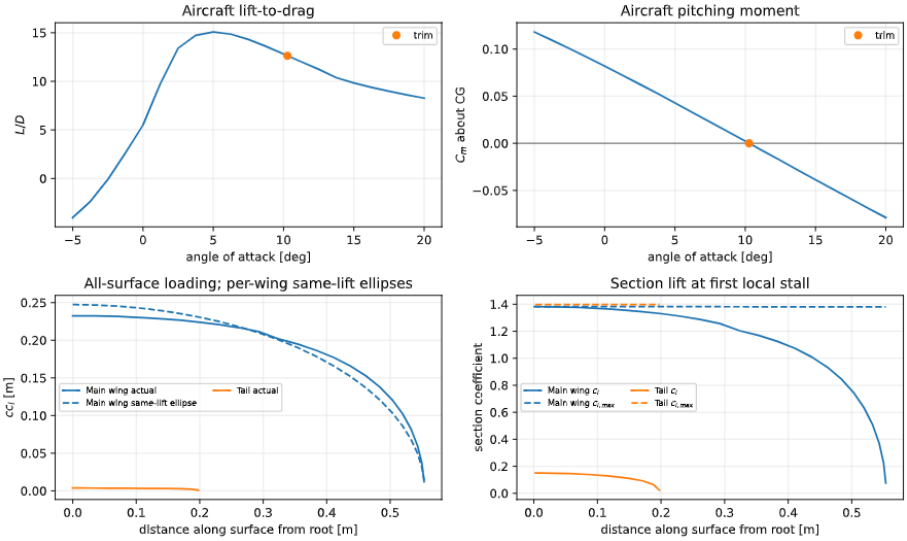


Figure 3. Genetic optimization results



With our optimized wing design, we have a minimum stall speed of around ~7 m/s, well within the range that RC trainer aircraft typically operate in (5 m/s - 8 m/s).